One policy. Two evidence channels. A tournament reward.

Policy-BT evaluates scientific outputs across workflow generations—not against hidden benchmark answers.

01 PRESERVE THE EVIDENCE

Workspace snapshots

Code, logs, tables
and visual artifacts

Reproduce → normalize

Required reproduction contract
Fresh-venv rerun by default
Schema adapters read real outputs

Ineligible generations receive 0

02 COMPARE UNDER ONE CONTRACT

Deterministic Python policy

Goal-defined criteria, fixed across the task
Earliest decisive stage → win / loss / tie

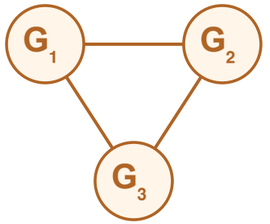
Only on a tie

Image-based judge

Relevant images on both sides
Agreement in both orders

Unresolved visual comparison remains tied

03 UPDATE THE EVOLUTION SIGNAL



All-pairs tournament

Bradley–Terry rewards

Fit strengths from pair outcomes
Refresh every current rating
Evidence → actionable feedback

Reward ≠ raw win rate ≠ benchmark grade

Selection + evidence-grounded mutation

Generate the next workspace

Code decides first. Images can break a tie. Bradley–Terry converts the verdict history into a current reward.